

Research Next Plan Note

Project: MedGemma4B skin disease multimodal ablation study
Prepared from Kaggle run review: 28 July 2026

Current Evidence

- Main multimodal model held-out test accuracy: 92.77%, macro F1: 93.42%, ROC-AUC: 0.9887.
- Ablation 1 visual-only held-out test accuracy: 89.16%, macro F1: 90.10%, ROC-AUC: 0.9849.
- Ablation 2 text-only held-out test accuracy: 19.28%, macro F1: 8.08%, ROC-AUC: 0.5000.
- Text-only model collapsed to one-class prediction, so there is no evidence of useful text-only signal or text-label leakage.
- Kaggle logs reported CV/test image path overlap = 0, exact image hash overlap = 0, and disease-name text leakage audit = PASS.

Next Planned Experiment

- Run a neutral-text control using the main model code.
- Keep the image branch, DinoV2 encoder, cross-attention/fusion structure, 5-fold training, and held-out test evaluation unchanged.
- Replace every sample text with the same neutral sentence, for example: "patient presents with observed cutaneous skin lesions and clinical symptoms."
- Suggested Kaggle title: Neutral Text Control Main Model T4x2.
- Use Kaggle GPU T4 x2.
- Estimated full training and test runtime: about 6.5 to 7.5 hours; allow up to 8 hours for cache/model loading delays.

Interpretation Rule

- If neutral-text control remains around 92-93% accuracy, the main model gain is likely from architecture/fusion behavior rather than meaningful sample-specific text.
- If neutral-text control drops near the visual-only result around 89% or lower, the original text stream likely provides a small complementary signal.
- If neutral-text control drops much lower than visual-only, the multimodal fusion may depend strongly on non-neutral text alignment and should be analyzed further.

Recommended Later Audits

- Patient-level split audit if patient identifiers are available.
- Perceptual-hash or embedding-similarity near-duplicate audit to catch resized/cropped duplicate images.
- Report wording: no evidence of text-label leakage or exact image duplicate leakage was found, but patient-level and near-duplicate audits are recommended.